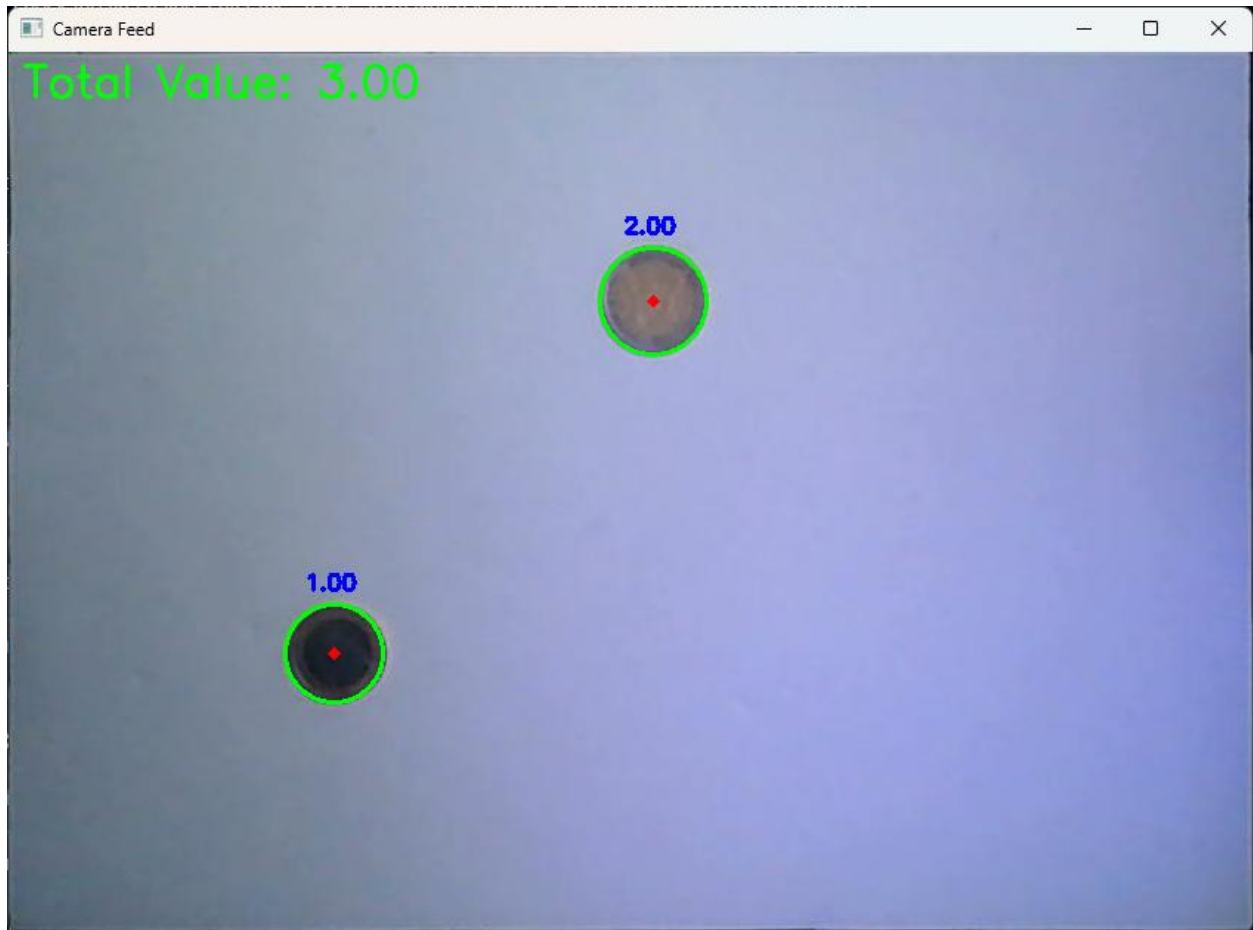


PaperCoin



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A computer vision application that automatically detects and counts Euro coins placed on an A4 paper using real-time camera feed. The system identifies coin denominations based on their size and calculates the total monetary value.

Overview

This project uses OpenCV and Python to create a robust coin counting system that can handle various lighting conditions and camera angles. By using an A4 paper as a reference surface, the system provides consistent detection regardless of camera positioning.

Core Functions

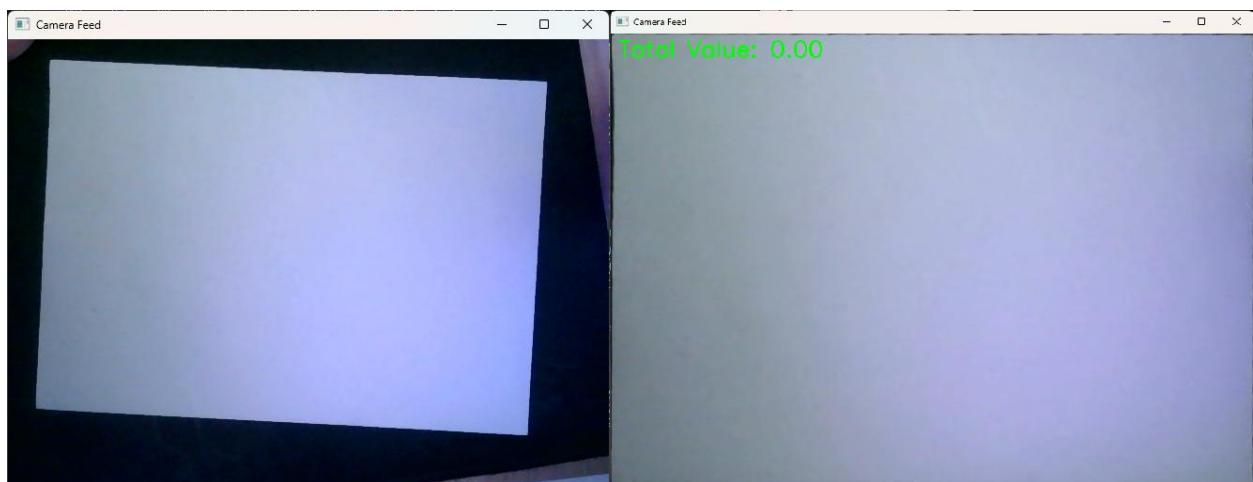
1. `find_and_crop_a4(frame)`

Purpose: Establishes a stable reference frame for coin detection.

This function locates an A4 paper in the camera view and performs perspective correction to create a flat, overhead view. The A4 paper serves as both a reference surface and a potential calibration target (though calibration is not yet fully implemented).

Key Operations:

- Converts frame to grayscale and applies Gaussian blur
- Uses Canny edge detection to find contours
- Identifies quadrilateral shapes with aspect ratio ~ 1.414 ($\sqrt{2}$, the A4 ratio)
- Applies perspective transform to "flatten" the paper
- Implements temporal smoothing to prevent jittery detection
- Automatically detects orientation (portrait vs. landscape)



2. `hough_circle_detection(coins, min_r, max_r)`

Purpose: Detects circular shapes (coins) in the processed image.

This is the heart of the coin detection system, using OpenCV's Hough Circle Transform to identify circular objects within specified radius ranges.

Key Operations:

- Converts to grayscale and applies median blur
- Runs Hough Circle Transform with tuned parameters
- Returns detected circles with center coordinates (x, y) and radius (r)



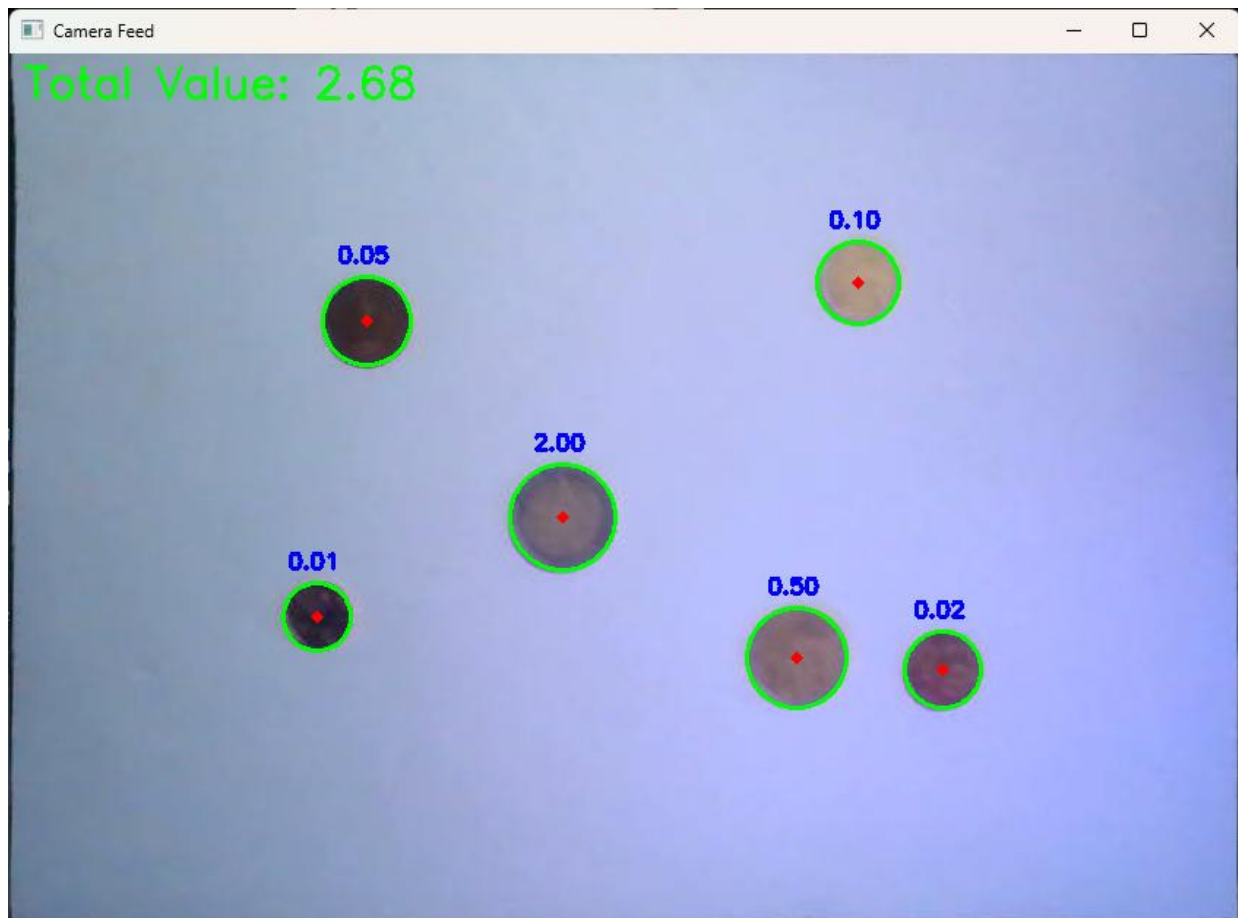
3. find_coins(frame)

Purpose: Identifies, tracks, and values each coin in the frame.

This function combines detection with intelligent tracking and filtering to provide stable, accurate coin counting.

Key Operations:

- Calls `hough_circle_detection()` to find potential coins
- **Coin Tracking:** Matches detected coins across frames using position-based IDs
- **Radius Filtering:** Maintains a 30-frame history of each coin's radius and uses the mean to reduce noise
- **Value Assignment:** Maps filtered radius to Euro denominations using the `COIN_VALUES` dictionary
- **Visual Output:** Draws circles and labels on each detected coin
- **Cleanup:** Removes coins from history when no longer detected



Methodologies

1. Contour Detection & Polygon Approximation

The A4 paper detection relies on finding the largest quadrilateral contour that matches the expected aspect ratio. This is achieved through:

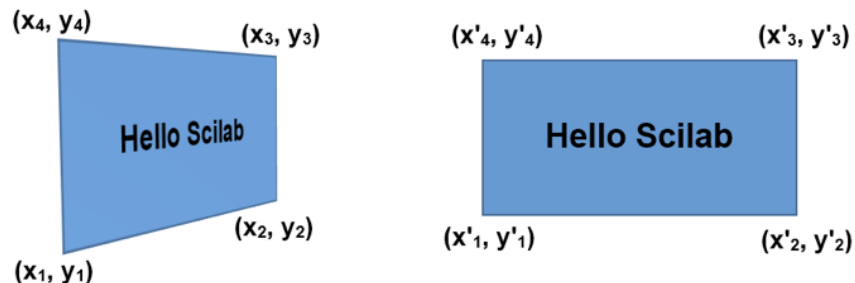
- **Canny Edge Detection:** Identifies edges in the image
- **findContours():** Extracts boundary shapes
- **approxPolyDP():** Simplifies contours into polygons
- **Aspect Ratio Filtering:** Ensures the detected shape matches A4 proportions



2. Perspective Transform

Once the four corners of the A4 paper are identified, a perspective transformation creates a "bird's eye view" of the surface. This involves:

- **Corner Ordering:** Sorting points using sum and difference operations to consistently identify top-left, top-right, bottom-right, and bottom-left
- **getPerspectiveTransform():** Calculates the transformation matrix
- **warpPerspective():** Applies the transformation to create a flat, rectangular view

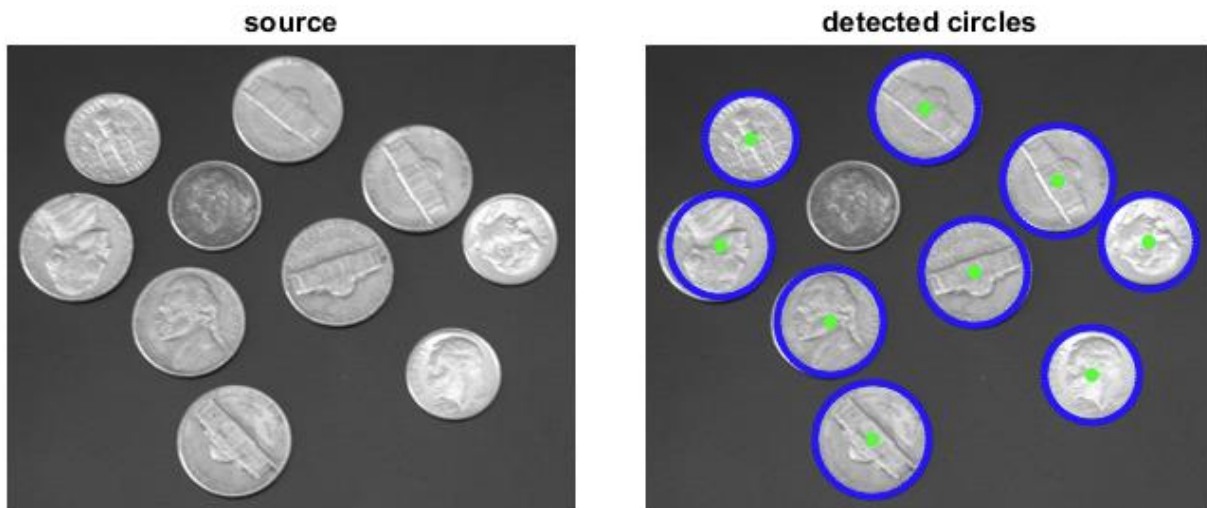


3. Hough Circle Transform

The Hough Circle Transform is a feature extraction technique specifically designed to detect circles in images. It works by:

- Detecting edge pixels (using Canny internally)
- For each edge pixel, considering all possible circles that could pass through it
- Accumulating "votes" in parameter space (center x, center y, radius)
- Identifying peaks in this accumulator as detected circles

This method is robust to incomplete circles and noise, making it ideal for coin detection.



4. Smoothing & Filtering

To prevent flickering and unstable detections, the system implements several smoothing techniques:

A4 Paper Smoothing:

- Tracks corner positions across frames
- Only updates position if movement exceeds a threshold
- Keeps previous position for small movements to reduce jitter

Coin Radius Filtering:

- Maintains a 30-frame rolling history of each coin's detected radius
- Uses the mean radius value for classification

- Reduces noise from slight detection variations

Coin Tracking:

- Assigns unique IDs to coins based on position
- Tracks coins across frames by matching nearest positions
- Preserves radius history even when coins move slightly

5. Radius-Based Classification

Euro coins have standardized diameters, so coins can be identified by measuring their size. The system:

- Detects coin radius in pixels
- Maps radius ranges to specific Euro denominations
- Uses averaged radius values for more accurate classification



Resources

Hough Circle Transform

- [Circle Detection using OpenCV - PyImageSearch](#) - Tutorial on Hough Circle detection
- [Hough Circle Transform Explained - YouTube](#) - Visual explanation of how the algorithm works
- [OpenCV Documentation - Hough Circle Transform](#) - Official documentation

Perspective Transform

- [Warp Perspective with OpenCV | Document Scanner](#) - Four-point perspective transform tutorial
- [Perspective Transformation - PyImageSearch](#) - Practical application of perspective transforms

Coin Detection Specific

- [Coin Detection using Python OpenCV - GeeksforGeeks](#) - Specific coin detection examples
- [Object Detection with OpenCV and Python - YouTube](#) - General object detection techniques

AI & Communities

- [Claude Sonnet 4.5](#) - AI assistant for code help and problem-solving
- [Stack Overflow](#) - Community Q&A from old issues
- [Reddit](#) – Other Projects and solved problems